

rumo ao

HTTP 2.0

*o que vem por aí e o que
você pode utilizar já com*

SPDY



Luiz Corte Real
@srsaude



Sérgio Lopes
@sergio_caelum



lançamento

Coletânea Front-end

Uma antologia da comunidade
front-end brasileira



ALMIR FILHO
BERNARD DE LUNA
CAIO GONDIM
DEVID MARQUES
DIEGO EIS
EDUARDO SHIOTA

GIOVANNI KEPPELEN
LUIZ CORTE REAL
JAYDSO GOMES
REINALDO FERRAZ
SÉRGIO LOPES

review

HTTP

em 5 slides

```
$ telnet www.caelum.com.br 80
```

```
Trying 64.233.171.121...
```

```
Connected to ghs.googlehosted.com.
```

```
Escape character is '^['.
```

```
GET / HTTP/1.1
```

```
Host: www.caelum.com.br
```

```
HTTP/1.1 200 OK
```

```
Content-Type: text/html; charset=utf-8
```

```
Vary: Accept-Encoding,User-Agent
```

```
Content-Language: pt-br
```

```
Date: Thu, 03 Apr 2014 18:37:18 GMT
```

```
Server: Google Frontend
```

```
Cache-Control: private
```

```
<!DOCTYPE html><html><head><title>Caelum</title>
```

```
$ telnet www.caelum.com.br 80
Trying 64.233.171.121...
Connected to ghs.googlehosted.com.
Escape character is '^]'.
```

```
GET / HTTP/1.1
```

TEXT0

```
Host: www.caelum.com.br
```

```
HTTP/1.1 200 OK
```

```
Content-Type: text/html; charset=utf-8
```

```
Vary: Accept-Encoding,User-Agent
```

```
Content-Language: pt-br
```

```
Date: Thu, 03 Apr 2014 18:37:18 GMT
```

```
Server: Google Frontend
```

```
Cache-Control: private
```

```
<!DOCTYPE html><html><head><title>Caelum</title>
```

```
$ telnet www.caelum.com.br 80
Trying 64.233.171.121...
Connected to ghs.googlehosted.com.
Escape character is '^]'.
```

```
GET / HTTP/1.1
Host: www.caelum.com.br
```

REQUEST

```
HTTP/1.1 200 OK
Content-Type: text/html; charset=utf-8
Vary: Accept-Encoding,User-Agent
Content-Language: pt-br
Date: Thu, 03 Apr 2014 18:37:18 GMT
Server: Google Frontend
Cache-Control: private
```

RESPONSE

```
<!DOCTYPE html><html><head><title>Caelum</title>
```

```
$ telnet www.caelum.com.br 80
Trying 64.233.171.121...
Connected to ghs.googlehosted.com.
Escape character is '^]'.
```

```
GET / HTTP/1.1
```

REQUEST

```
Host: www.caelum.com.br
```

... ESPERA ...

```
HTTP/1.1 200 OK
```

RESPONSE

```
Content-Type: text/html; charset=utf-8
```

```
Vary: Accept-Encoding,User-Agent
```

```
Content-Language: pt-br
```

```
Date: Thu, 03 Apr 2014 18:37:18 GMT
```

```
Server: Google Frontend
```

```
Cache-Control: private
```

```
<!DOCTYPE html><html><head><title>Caelum</title>
```

```
$ telnet www.caelum.com.br 80
Trying 64.233.171.121...
Connected to ghs.googlehosted.com.
Escape character is '^]'.
```

GET / HTTP/1.1 MÉTODO E URL

Host: www.caelum.com.br

HTTP/1.1 200 OK STATUS

Content-Type: text/html; charset=utf-8

Vary: Accept-Encoding, User-Agent

Content-Language: pt-br

Date: Thu, 03 Apr 2014 18:37:18 GMT

Server: Google Frontend

Cache-Control: private

RESPOSTA

<!DOCTYPE html><html><head><title>Caelum</title>

```
$ telnet www.caelum.com.br 80
Trying 64.233.171.121...
Connected to ghs.googlehosted.com.
Escape character is '^]'.
```

```
GET / HTTP/1.1
```

```
Host: www.caelum.com.br
```

HEADERS

```
HTTP/1.1 200 OK
```

```
Content-Type: text/html; charset=utf-8
```

```
Vary: Accept-Encoding,User-Agent
```

```
Content-Language: pt-br
```

```
Date: Thu, 03 Apr 2014 18:37:18 GMT
```

```
Server: Google Frontend
```

```
Cache-Control: private
```

HEADERS

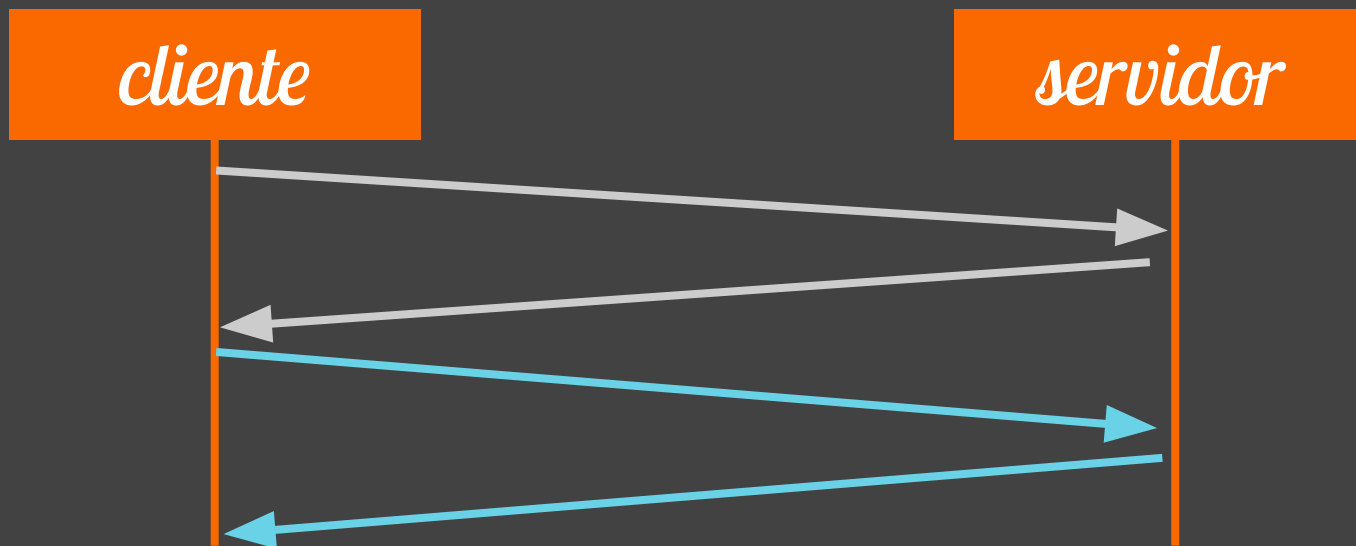
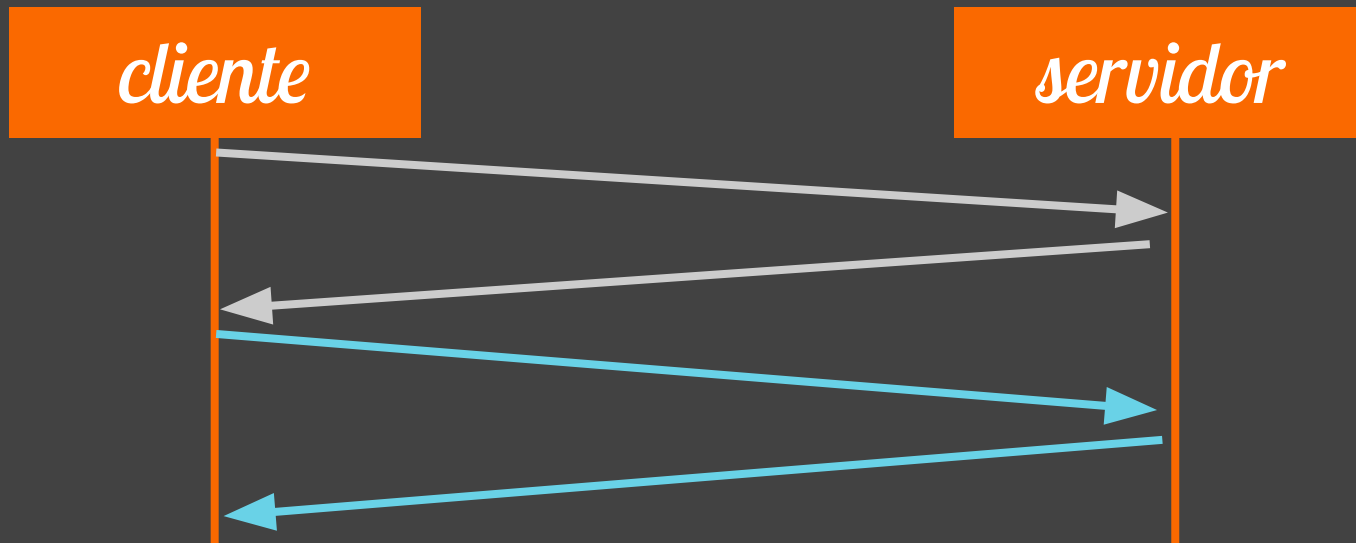
```
<!DOCTYPE html><html><head><title>Caelum</title>
```

http

texto

headers

request > espera > response



HTTP 1.1

cliente

servidor

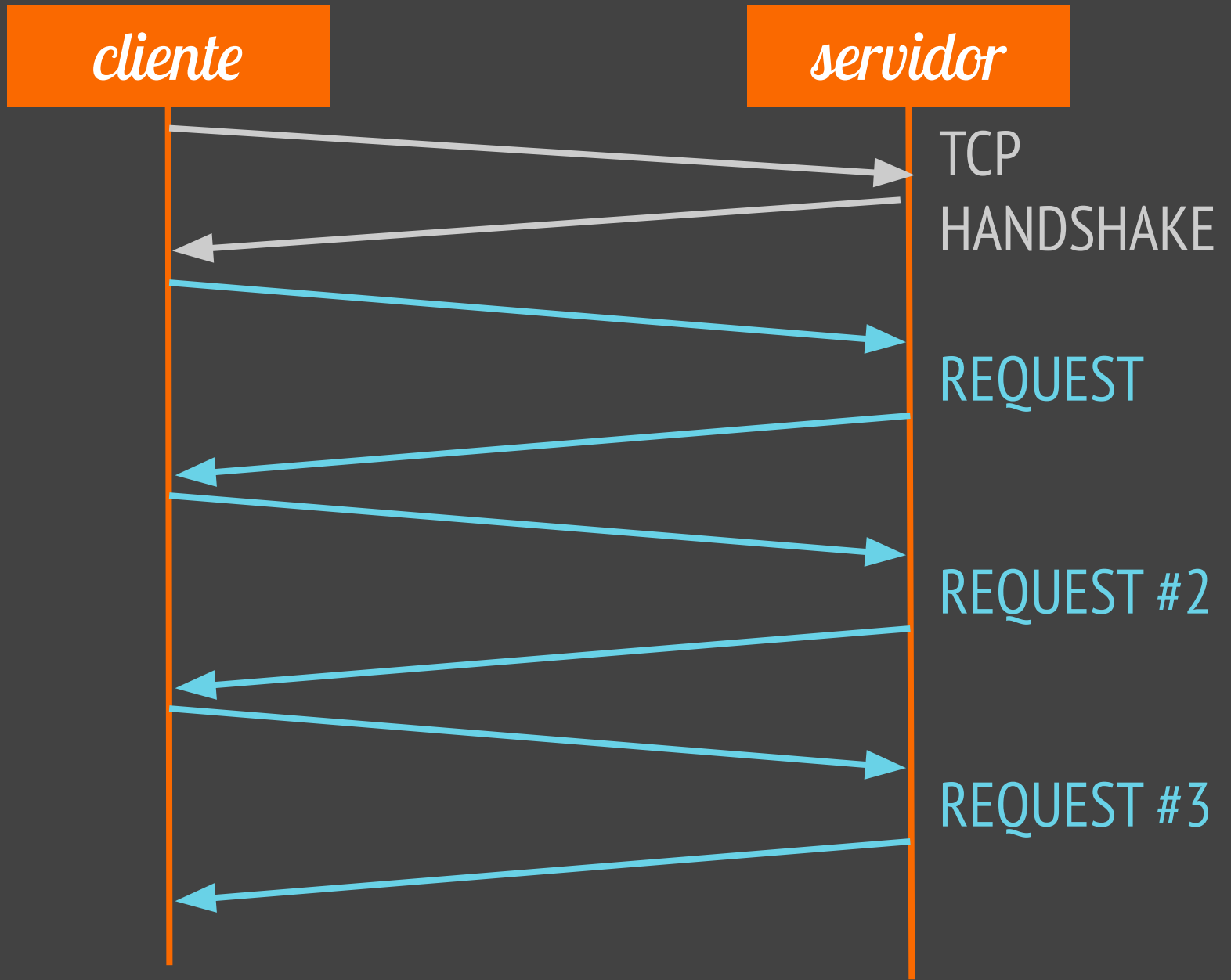
TCP
HANDSHAKE

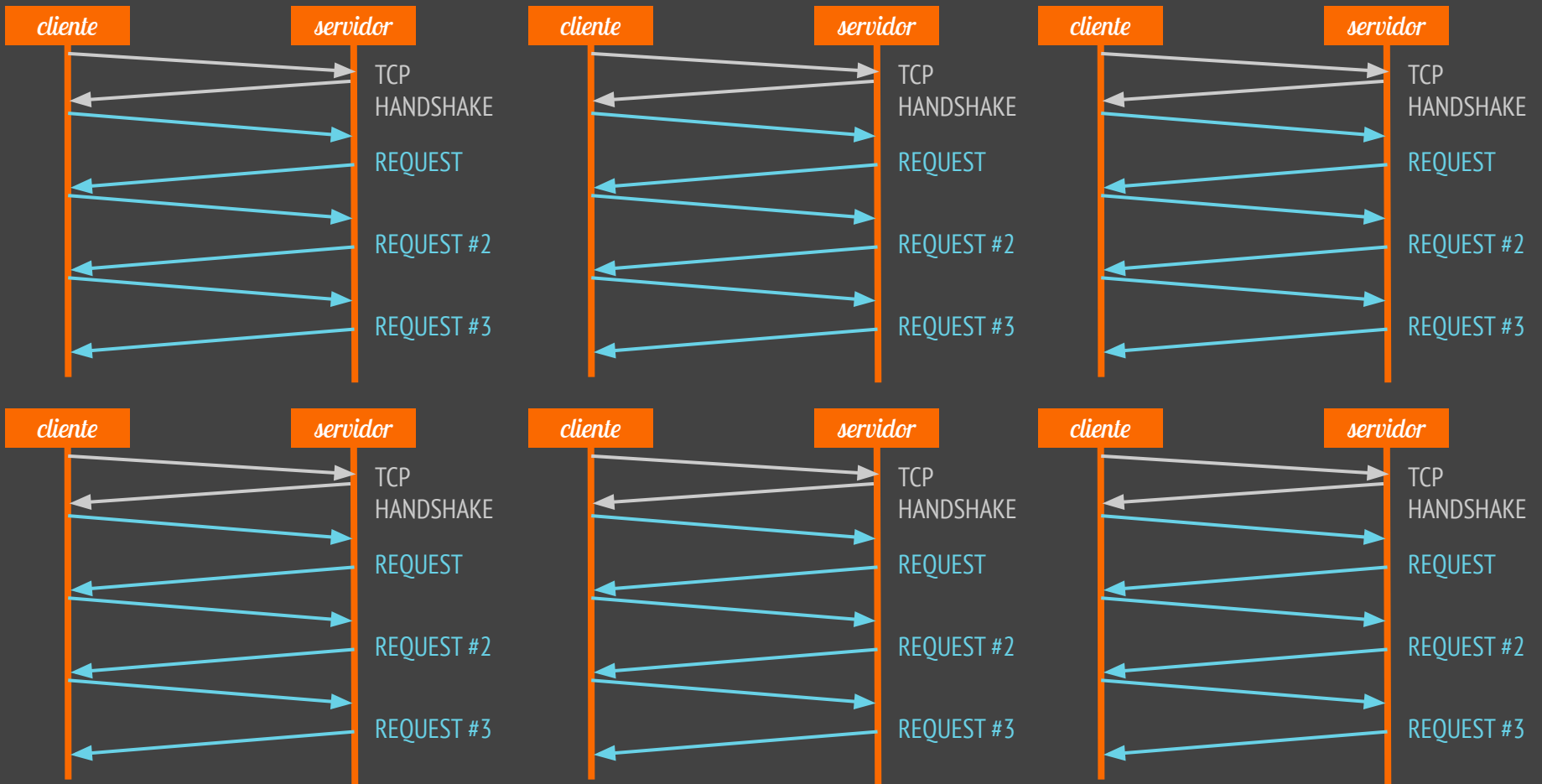
REQUEST

REQUEST #2

REQUEST #3

Keep-Alive





de 4 a 8 conexões

otimizações web

diminuir
requests



JUNTAR CSS E JS
SPRITES
INLINE DE RECURSOS

diminuir
tráfego



GZIP
MINIFICAÇÃO
DOMÍNIOS SEM COOKIES

paralelizar
requests



MÚLTIPLOS HOSTNAMES
CDNs

o fantástico mundo do

HTTP 2.0

(e do SPDY)

GET / HTTP/1.1

Host: www.caelum.com.br

http 1.1

HTTP/1.1 200 OK

Content-Type: text/html; charset=utf-8

Content-Language: pt-br

Date: Thu, 03 Apr 2014 18:37:18 GMT

Cache-Control: private

```
<!DOCTYPE html><html><head><title>Caelum</title>  
<meta name=viewport content=width=device-width>  
<meta name=description content="A Caelum tem os cursos de  
Java, Android, Scrum, Rails, .NET, HTML e CSS mais  
reconhecidos no mercado"><style>
```

GET / HTTP/1.1

Host: www.caelum.com.br

Accept-Encoding: gzip

http 1.1

HTTP/1.1 200 OK

Content-Type: text/html; charset=utf-8

Content-Language: pt-br

Date: Thu, 03 Apr 2014 18:37:18 GMT

Cache-Control: private

Content-Encoding: gzip

Vary: Accept-Encoding

??[mŚ???,1?T?%.?JI?;1?R0??E?

д?H{{?un??Tm??:?U?5DI??su?L??!???

Üm?_?\{-??bΦ?u??6?1,w??N?C.?0?f*??□??BY?????1#m|?x???L?

Q?????U?.?MZ躔?M?JN_?I??RNN????}Y?U????8?

Π???id?c?8)E8??%:?\$D*□?????4????

((R2[??0?;s??ŕy\??Fd18.HK?x?-?\$???İ??????????0

GZIP

http 2.0

01010101010101010101010101

010101010101
0101010101010
101010101010
0101010101010

BINÁRIO

??[mŚ???,1?T?%.?JI?;1?R0??E?
д?H{{?un??Tm??:?U?5DI??su?L??!???
Üm?_?\{-??bΦ?u??6?1,w??N?C.?0?f*??□??Pv?????l#m|?x???L?
Q????U?.?MZ跚?M?JN_?I??RNN????}Y?U????8?
Π???id?c?8)E8??%:?\$D*□?????4?????
((R2[??0?;s???y\??Fd18.HK?x?-?\$???İİ??????????0

GZIP

http 2.0

?c?8)E8??%:?\$

?T?%.?JI?;1?R
,1?T?%.?JI?;1?R
?c?8)E8??%:?\$

BINÁRIO

HPACK

??[mŚ???,1?T?%.?JI?;1?R0??E?

д?H{{?un??Tm??:?U?5DI??su?L??!???

Üm?_?\{-??bΦ?u??6?1,w??N?C.?0?f*??□??Pv?????l#m|?x??L?

Q???'?U?.?MZ跚?M?JN_?I??RNN???'Y?U????8?

Π???id?c?8)E8??%:?\$D*□?????\$4???

((R2[??0?;s???y\??Fd18.HK?x?-?\$???İİ??????0

GZIP





ou não #heartbleed

??????????

???????

?? ? ? ? ? ? ? ?

????????

BINÁRIO

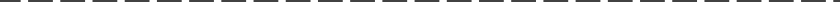
HPACK

SSL

[illegible]

?

CCIP CCIP



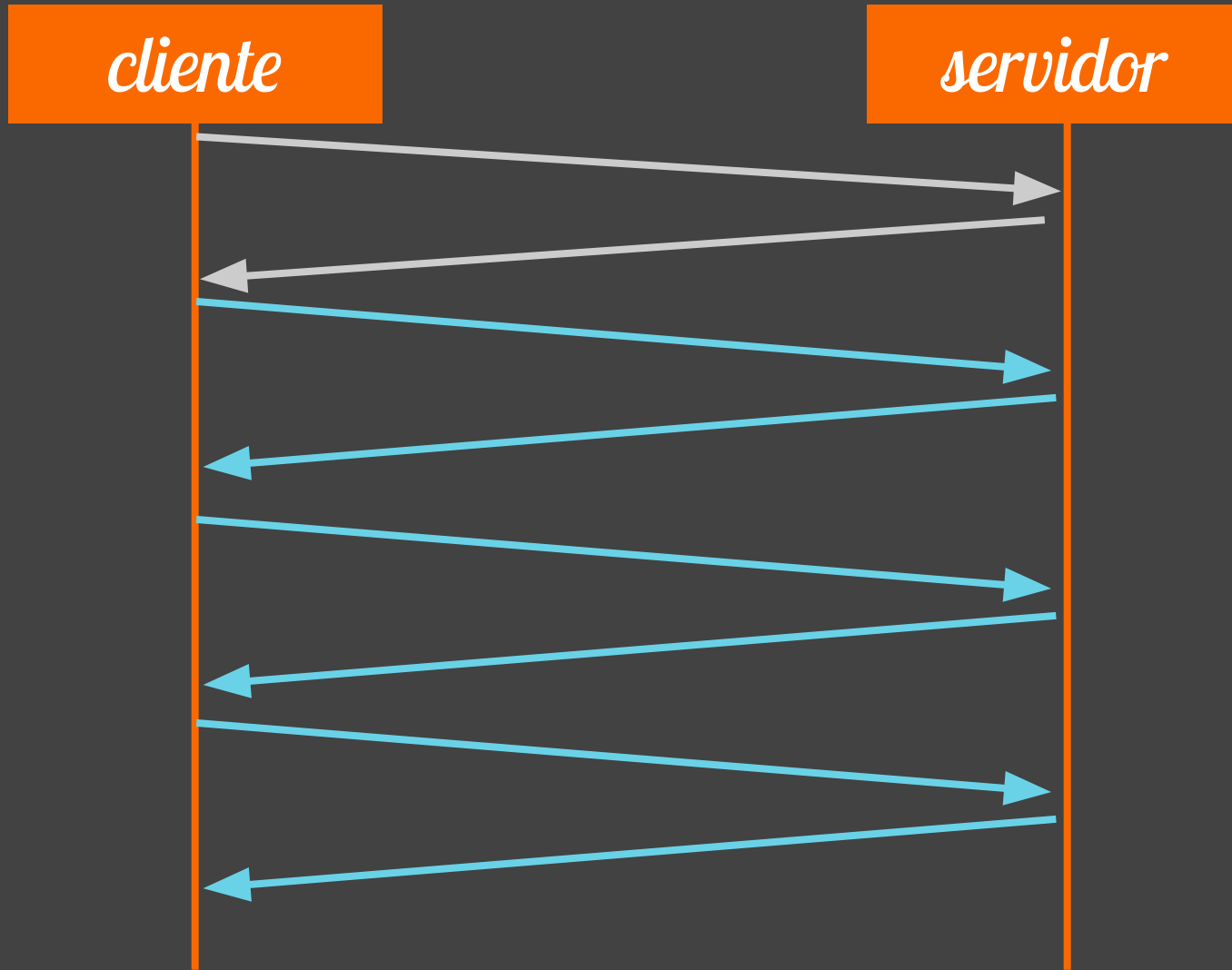


?

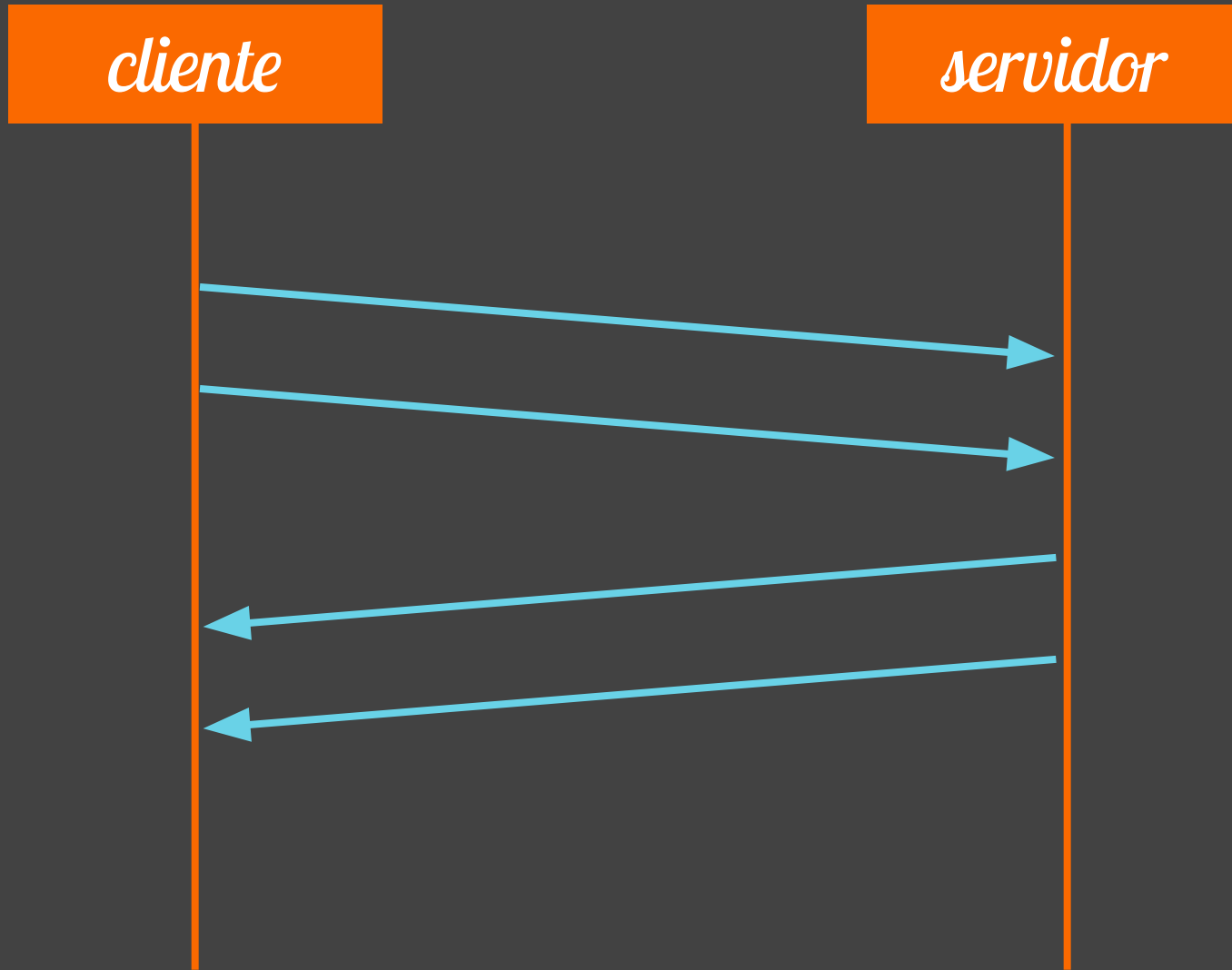
GZIP

SSL

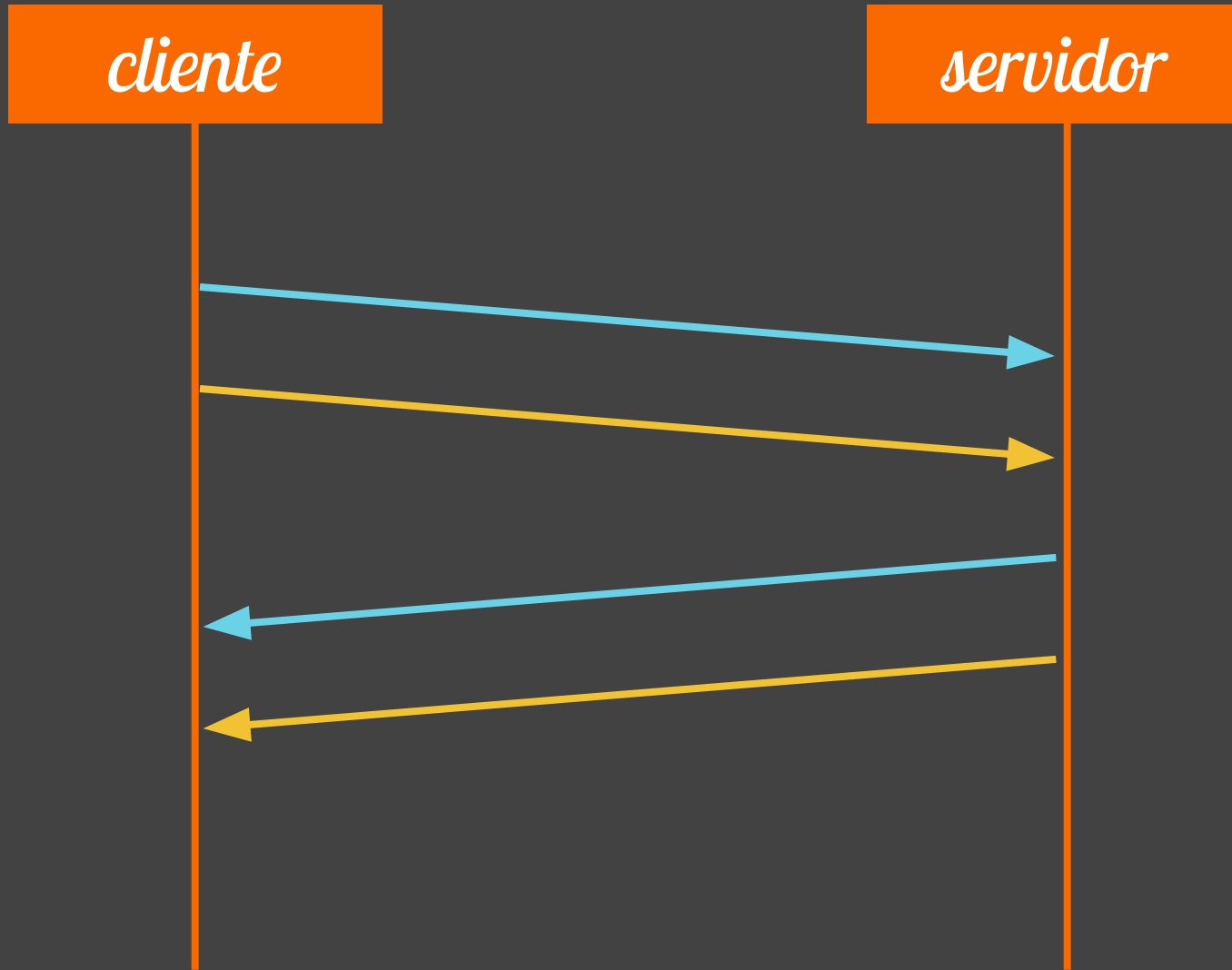
HEADER BINÁRIO
GZIP / HPACK
SSL



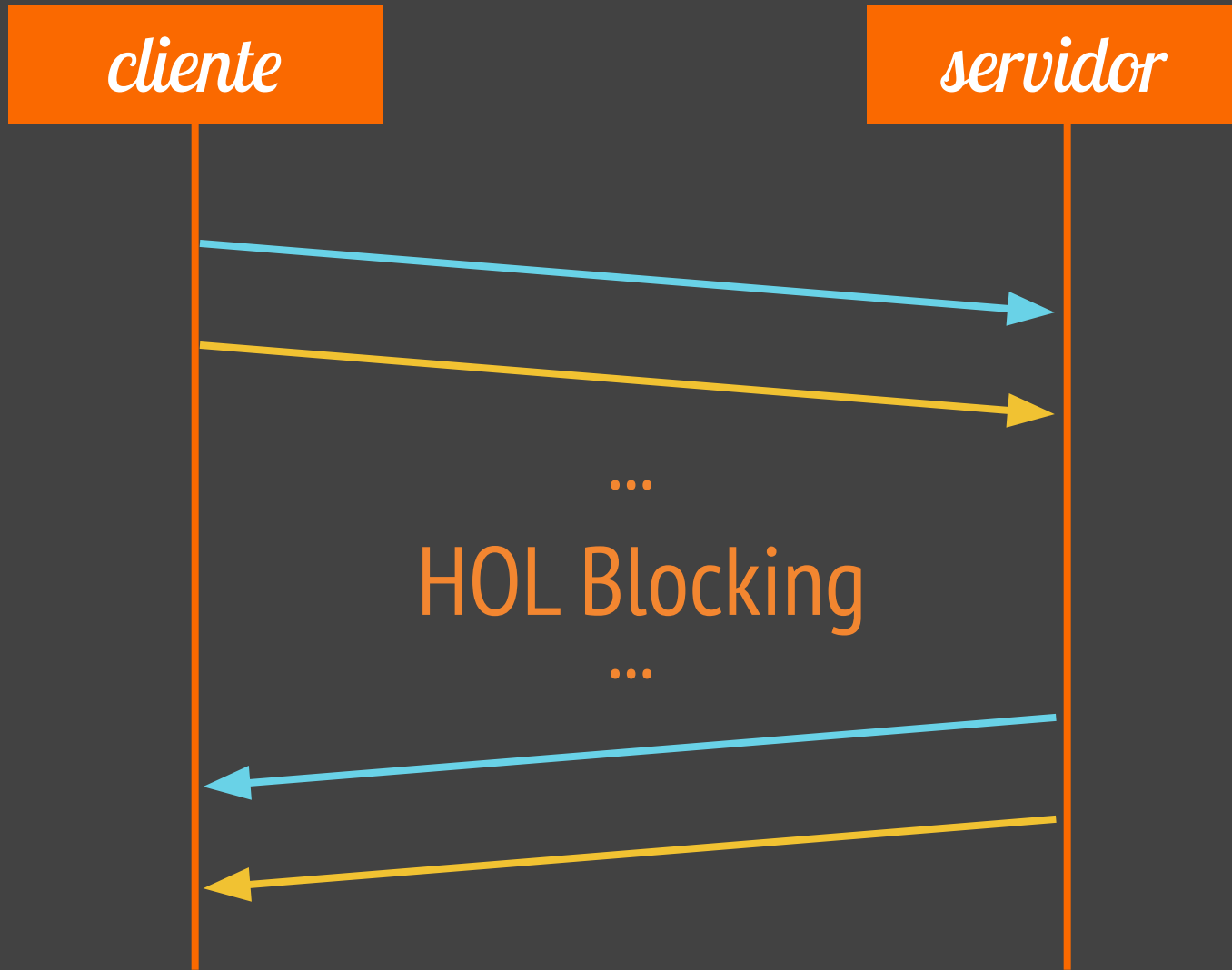
HTTP 1.1 com Keep-Alive



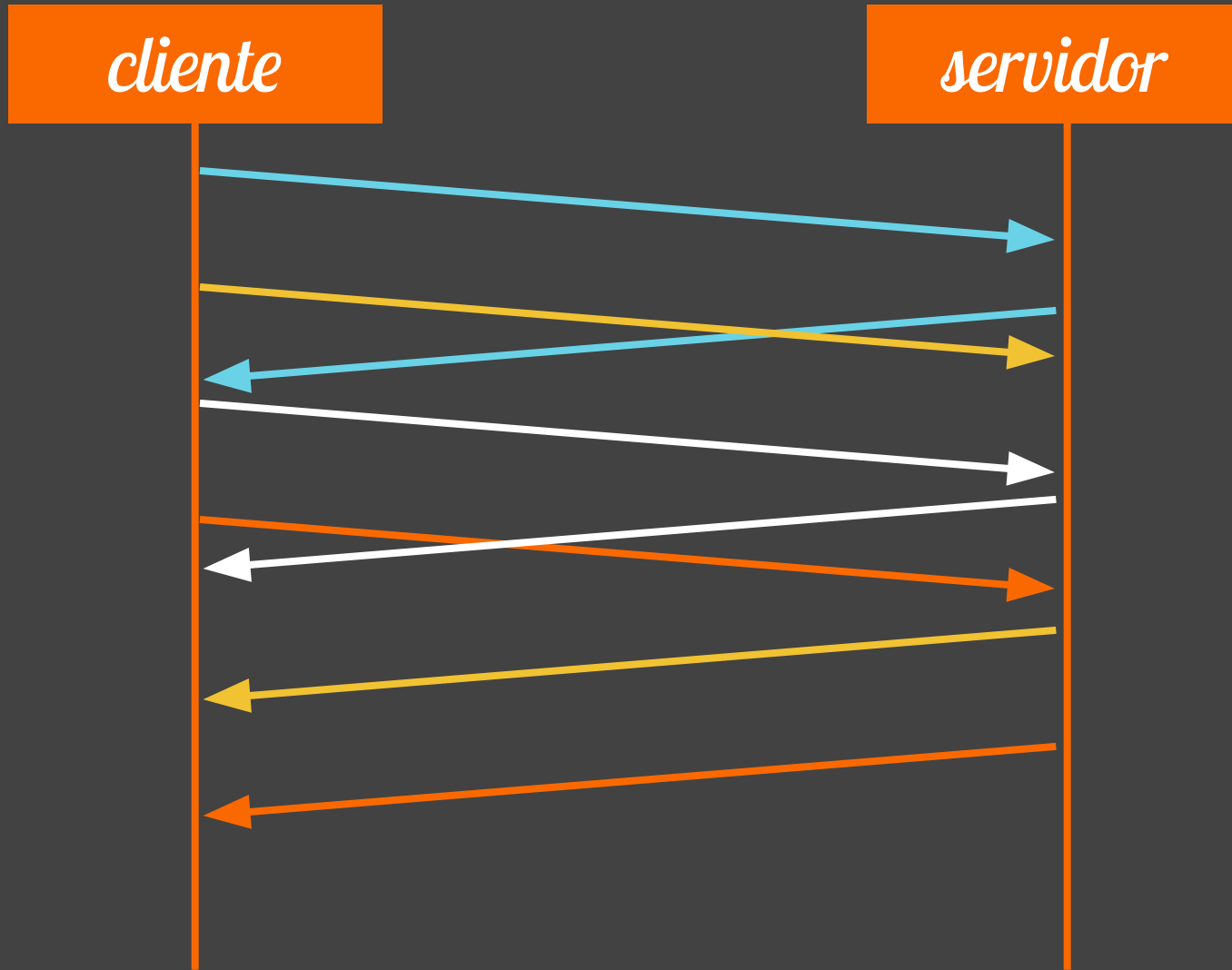
HTTP 1.1 com Pipelining*



HTTP 1.1 com Pipelining*



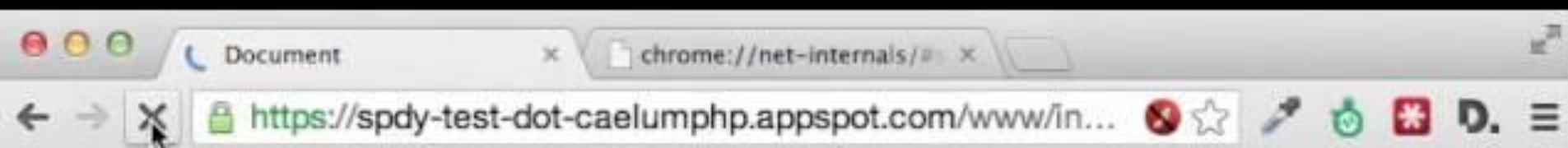
HTTP 1.1 com pipelining*



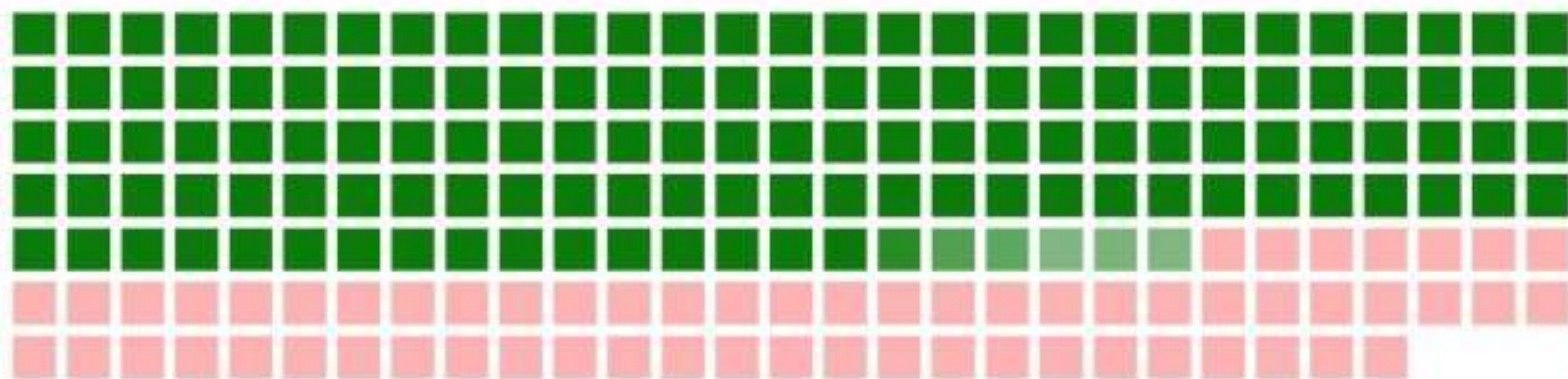
HTTP 2.0 multiplexing

multiplexing

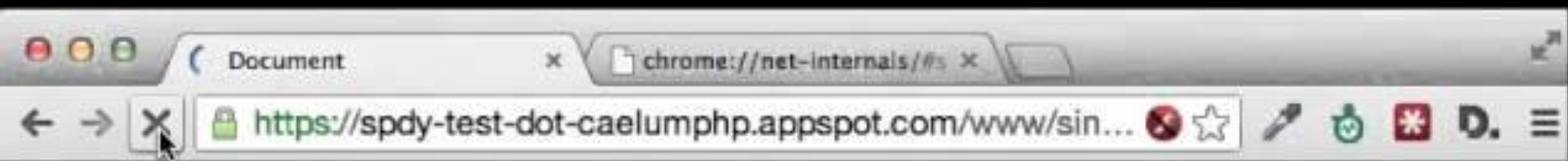
1 CONEXÃO TCP
ASSÍNCRONO



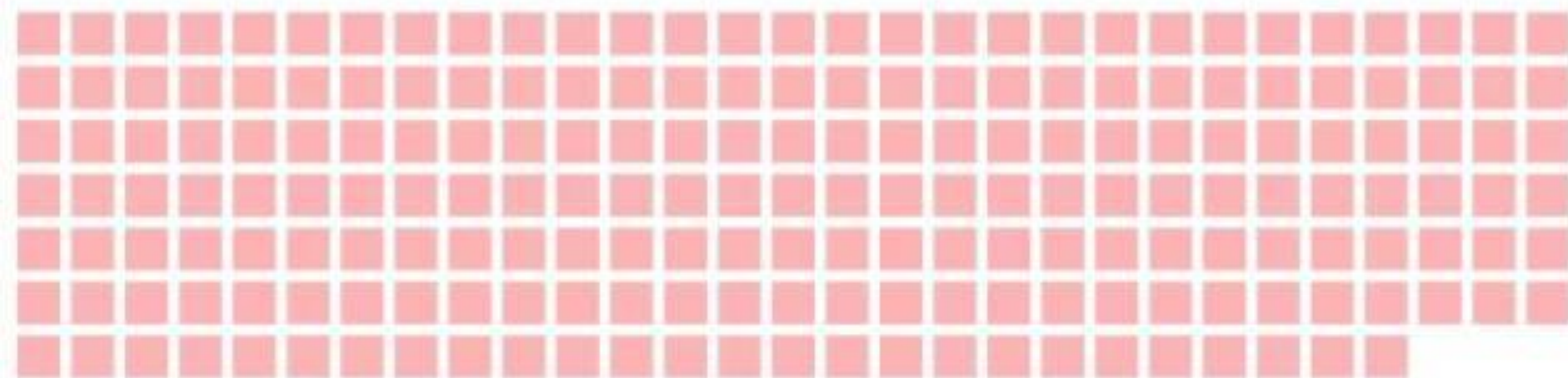
Loading 200 JS modules asynchronously in separated files (13 kb each file)

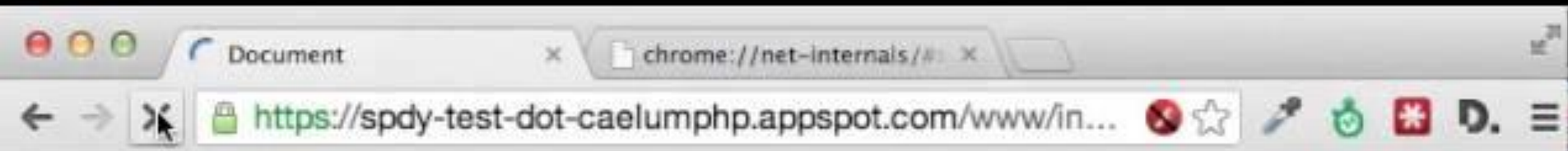


First script time: 1269 ms

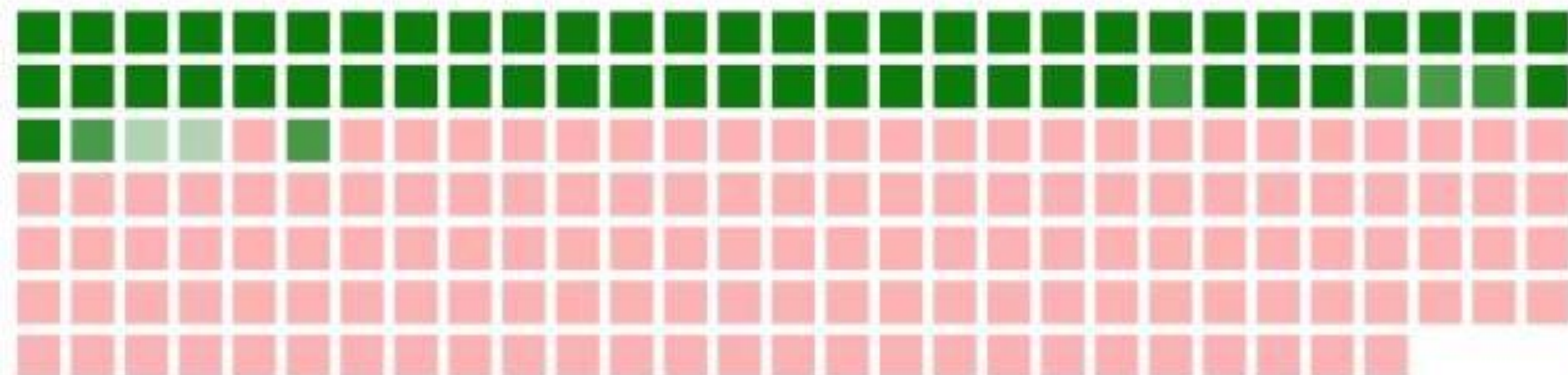


Loading 200 JS modules in one file (2 MB)





Loading 200 JS modules asynchronously in separated files (13 kb each file)



First script time: 1225 ms

otimizações web

diminuir
requests



JUNTAR CSS E JS
SPRITES
INLINE DE RECURSOS

diminuir
tráfego



GZIP
MINIFICAÇÃO
DOMÍNIOS SEM COOKIES

paralelizar
requests



MÚLTIPLOS HOSTNAMES
CDNs

otimizações web

diminuir
requests



JUNTAR CSS E JS
SPRITES
INLINE DE RECURSOS

diminuir
tráfego



GZIP
MINIFICAÇÃO
DOMÍNIOS SEM COOKIES

paralelizar
requests



MÚLTIPLOS HOSTNAMES
CDNs

otimizações web

diminuir
requests



JUNTAR CSS E JS
SPRITES
INLINE DE RECURSOS

diminuir
tráfego



GZIP ★
MINIFICAÇÃO
DOMÍNIOS SEM COOKIES

paralelizar
requests



MÚLTIPLOS HOSTNAMES
CDNs

otimizações web

diminuir
requests



JUNTAR CSS E JS
SPRITES
INLINE DE RECURSOS

diminuir
tráfego



GZIP ★
MINIFICAÇÃO
DOMÍNIOS SEM COOKIES

paralelizar
requests



MÚLTIPLOS HOSTNAMES
CDNs ★

http 1.1

:method: GET

:path: /

Host: www.caelum.com.br

Accept: text/html,application/xhtml+xml;q=0.9,image/jpeg

Accept-Encoding: gzip

User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_2)

Cookie: SID=786dads78asdbad876asdhjvb28

:method: GET

:path: /style.css

Host: www.caelum.com.br

Accept: text/html,application/xhtml+xml;q=0.9,image/jpeg

Accept-Encoding: gzip

User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_2)

Cookie: SID=786dads78asdbad876asdhjvb28

http 1.1

:method: GET

:path: /

Host: www.caelum.com.br

Accept: text/html,application/xhtml+xml;q=0.9,image/jpeg

Accept-Encoding: gzip

User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_2)

Cookie: SID=786dads78asdbad876asdhjvb28

:method: GET

:path: /style.css

Host: www.caelum.com.br

Accept: text/html,application/xhtml+xml;q=0.9,image/jpeg

Accept-Encoding: gzip

User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_2)

Cookie: SID=786dads78asdbad876asdhjvb28

http 2.0

:method: GET

:path: /

Host: www.caelum.com.br

Accept: text/html,application/xhtml+xml;q=0.9,image/jpeg

Accept-Encoding: gzip

User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_2)

Cookie: SID=786dads78asdbad876asdhjvb28

:path: /style.css

http 2.0

:method: GET

:path: /

Host: www.caelum.com.br

Accept: text/html,application/xhtml+xml;q=0.9,image/jpeg

Accept-Encoding: gzip

User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_2)

Cookie: SID=786dads78asdbad876asdhjvb28

:path: /style.css

:path: /main.js

:path: /imagem.jpg

Host: images.caelum.com.br

header tables

MENOS BYTES

REQUEST e RESPONSE

STATEFUL

otimizações web

diminuir
requests



JUNTAR CSS E JS
SPRITES
INLINE DE RECURSOS

diminuir
tráfego



GZIP ★
MINIFICAÇÃO
DOMÍNIOS SEM COOKIES

paralelizar
requests



MÚLTIPLOS HOSTNAMES
CDNs ★

otimizações web

diminuir
requests



JUNTAR CSS E JS
SPRITES
INLINE DE RECURSOS

diminuir
tráfego



GZIP ★
MINIFICAÇÃO
DOMÍNIOS SEM COOKIES

paralelizar
requests



MÚLTIPLOS HOSTNAMES
CDNs ★

inline

~~diminuir requests~~

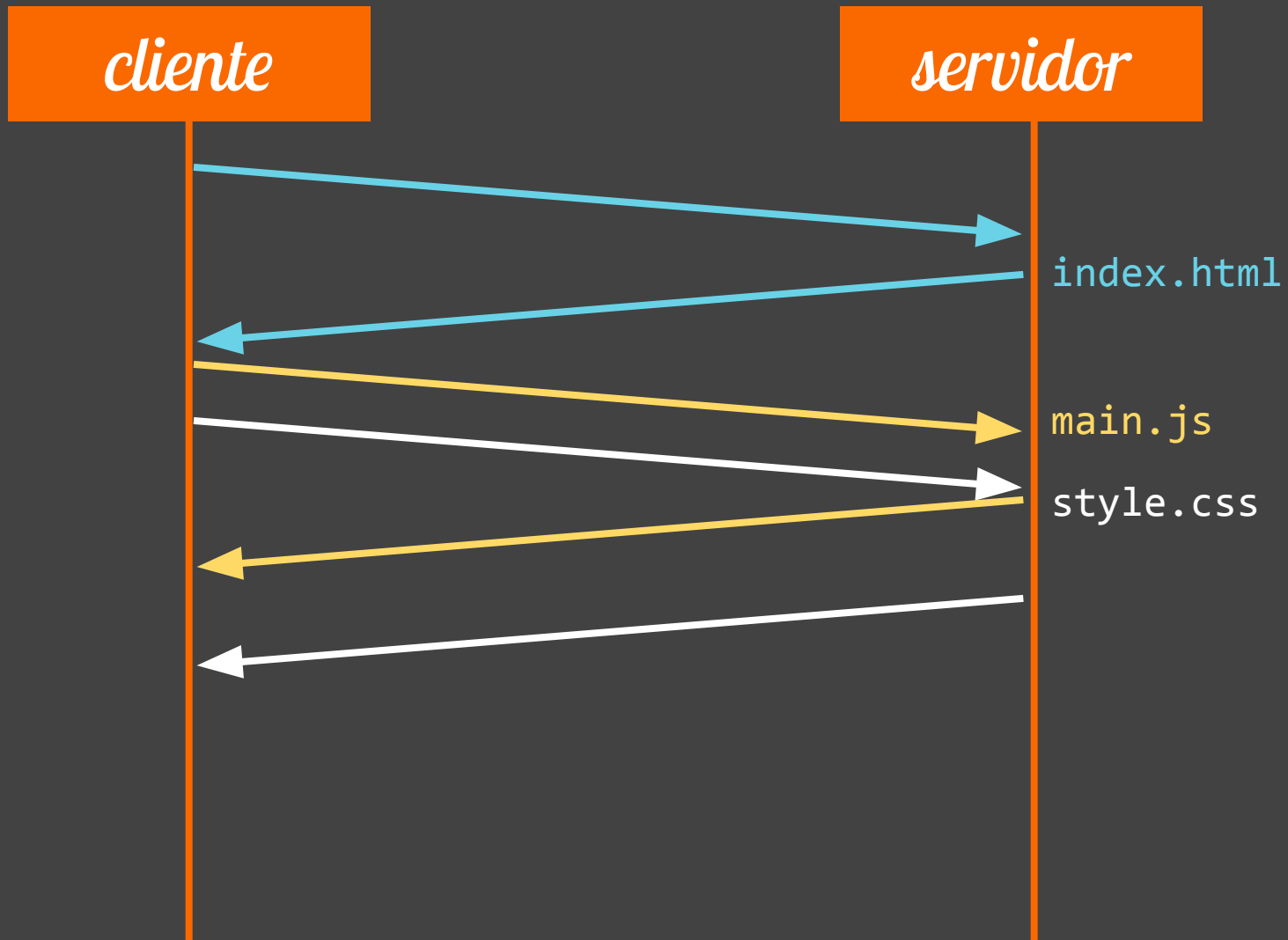
priorizar conteúdo

```
<!DOCTYPE html>
<html>
<head>
  <style>
    /* estilo para o ATF */
  </style>

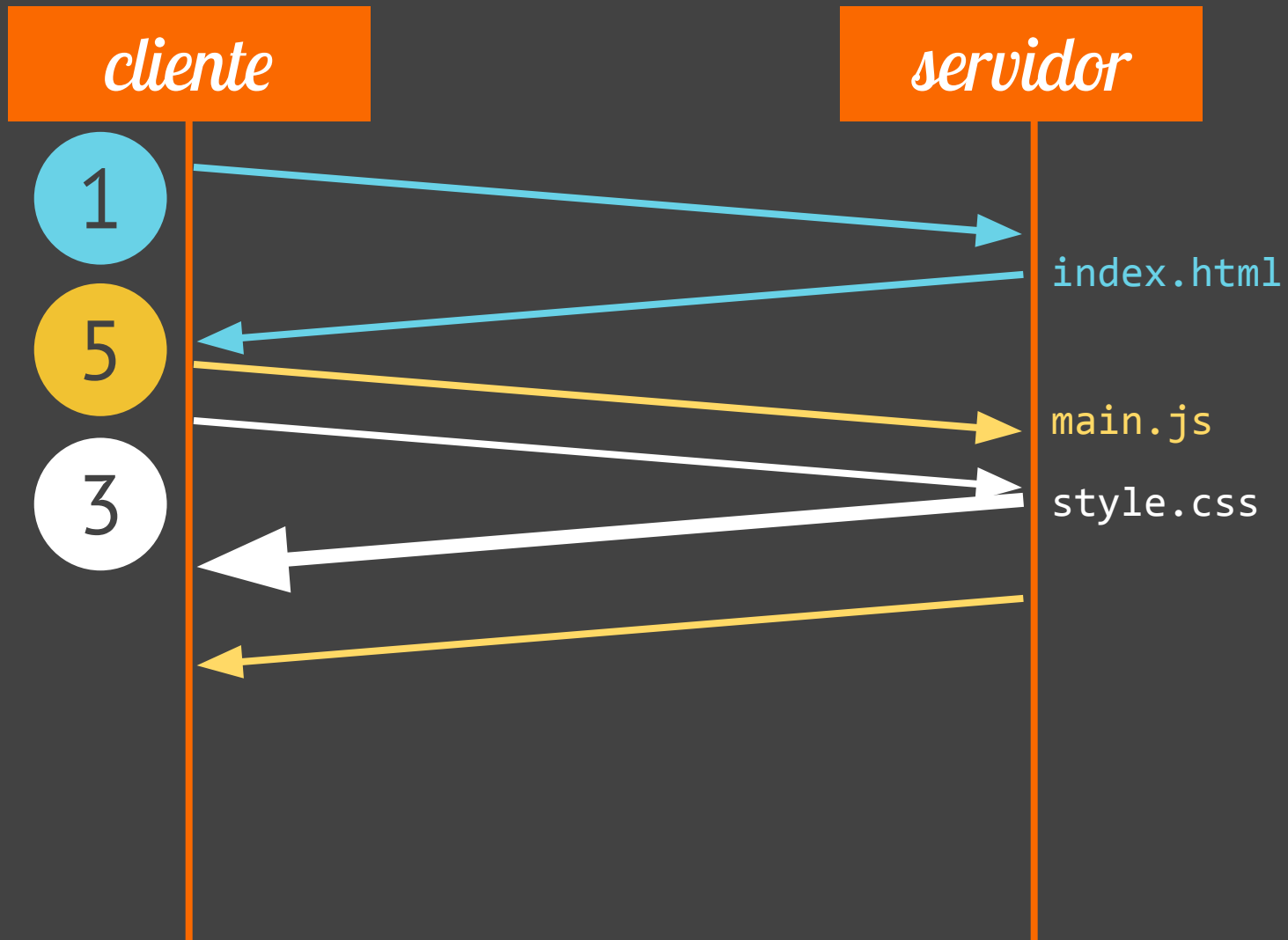
  <script>
    /* javascript importante */
  </script>

  /* javascript pra depois */
  <script src="secundario.js" async></script>
</head>
<body>
  ...
</body>
</html>
```

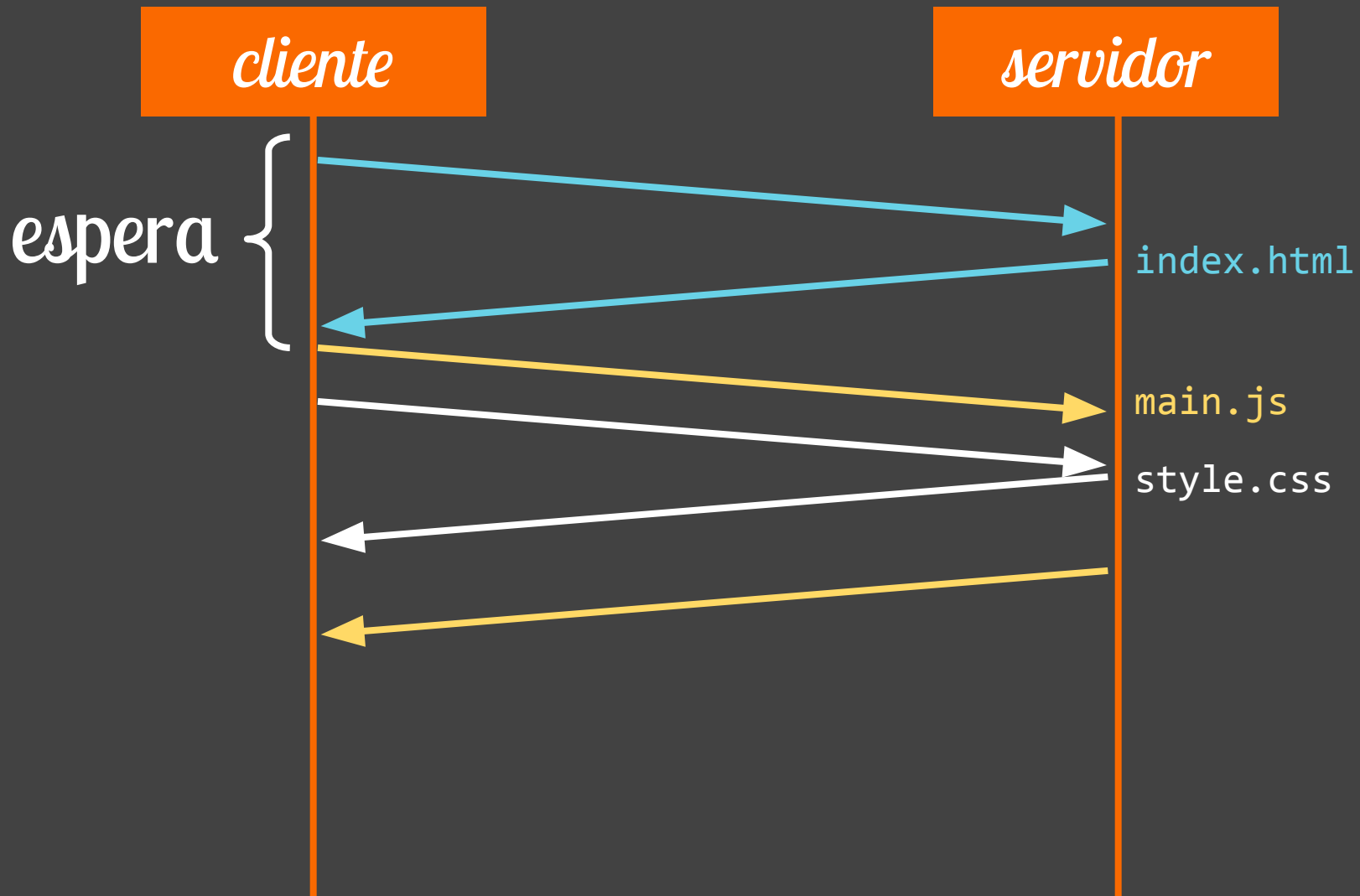
priorizar conteúdo



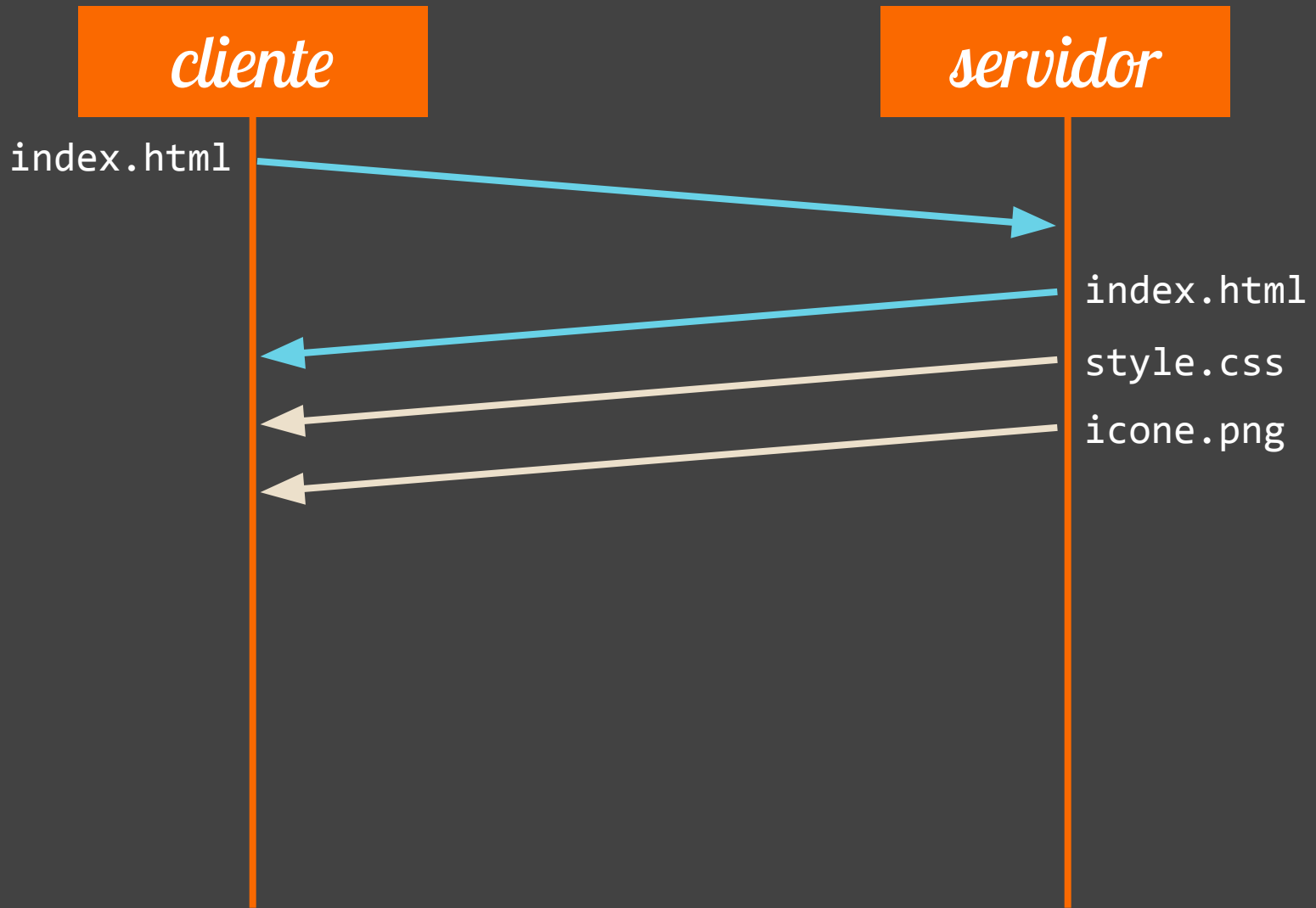
priorizar conteúdo



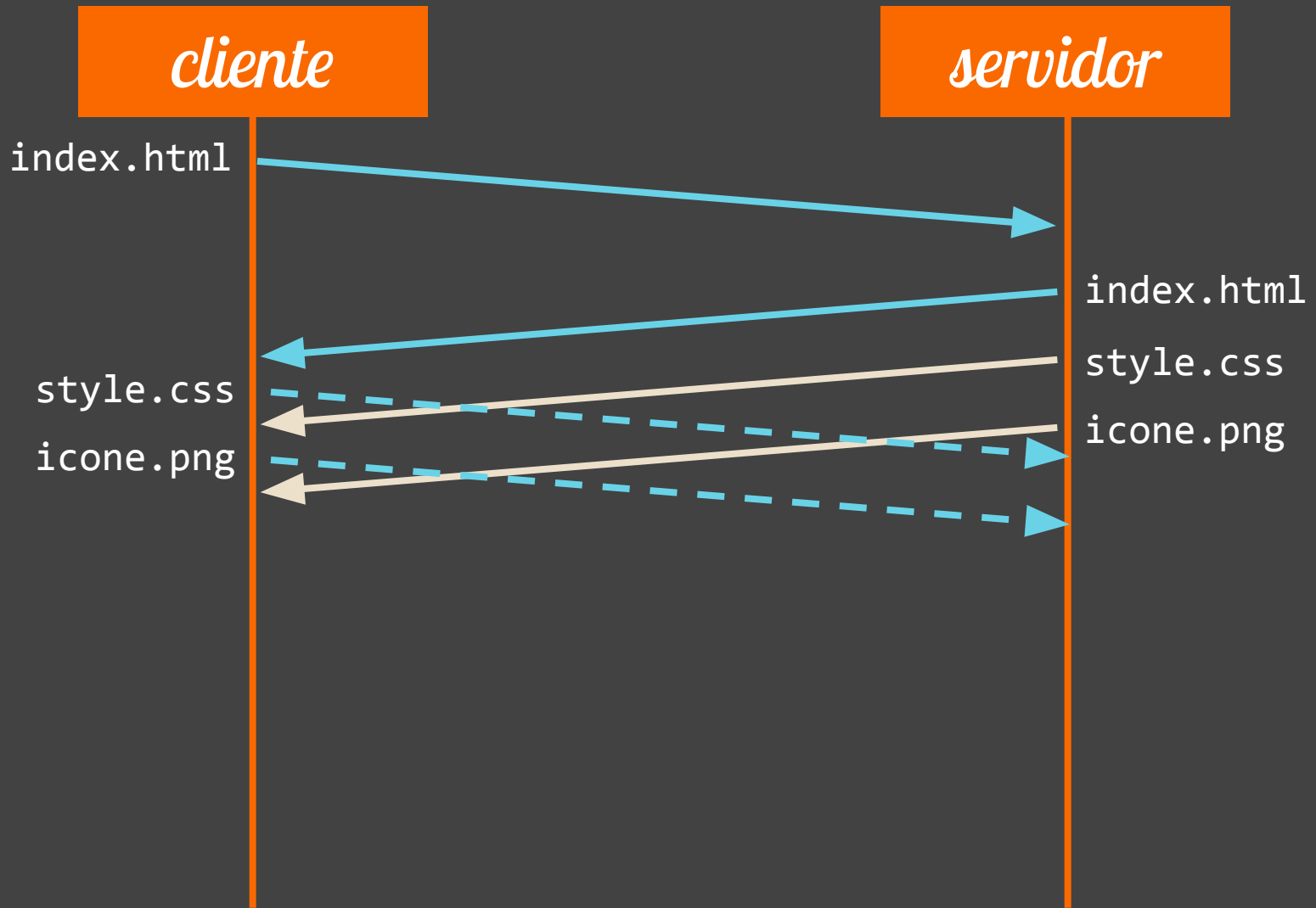
priorizar conteúdo



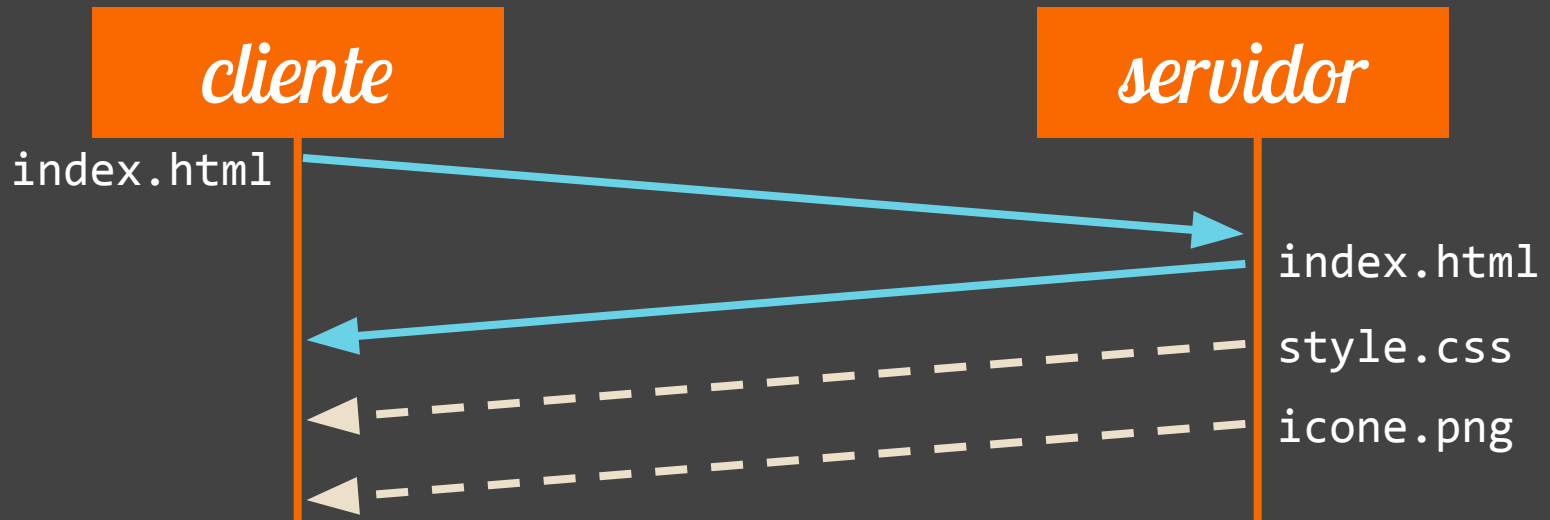
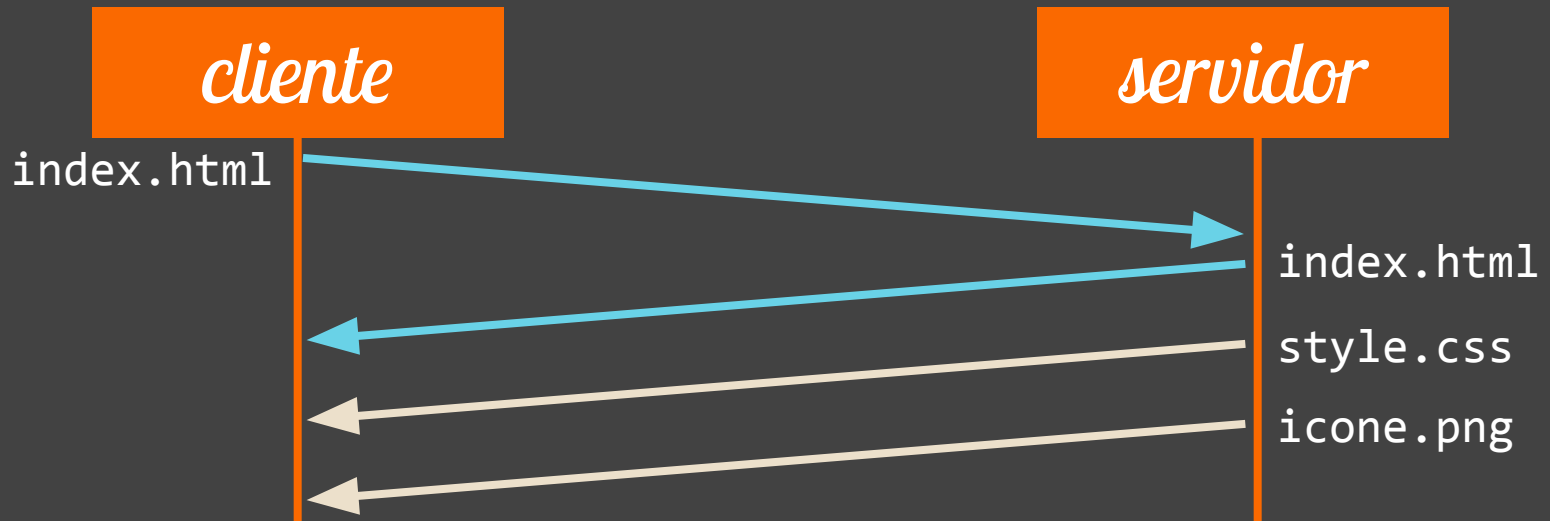
index.html bloqueante



Server Push



Server Push



Server Push cancelável

server push

PRIORIZAÇÃO

CANCELÁVEL

CACHEÁVEL

otimizações web

diminuir
requests



JUNTAR CSS E JS
SPRITES
INLINE DE RECURSOS

diminuir
tráfego



GZIP ★
MINIFICAÇÃO
DOMÍNIOS SEM COOKIES

paralelizar
requests



MÚLTIPLOS HOSTNAMES
CDNs ★

otimizações web

diminuir
requests



JUNTAR CSS E JS
SPRITES
INLINE DE RECURSOS

diminuir
tráfego



GZIP ★
MINIFICAÇÃO
DOMÍNIOS SEM COOKIES

paralelizar
requests



MÚLTIPLOS HOSTNAMES
CDNs ★

COMPRESSÃO MULTIPLEXING

CRİPTOGRAFIA PRIORIZAÇÃO

BİNÁRIO SERVER PUSH

http 2

+PERFORMANCE +FÁCİL +LEVE

+MOBILE +COMPATÍVEL +SEGURO

spdy

Google:
até 45% mais rápido

spdy

Real: depende



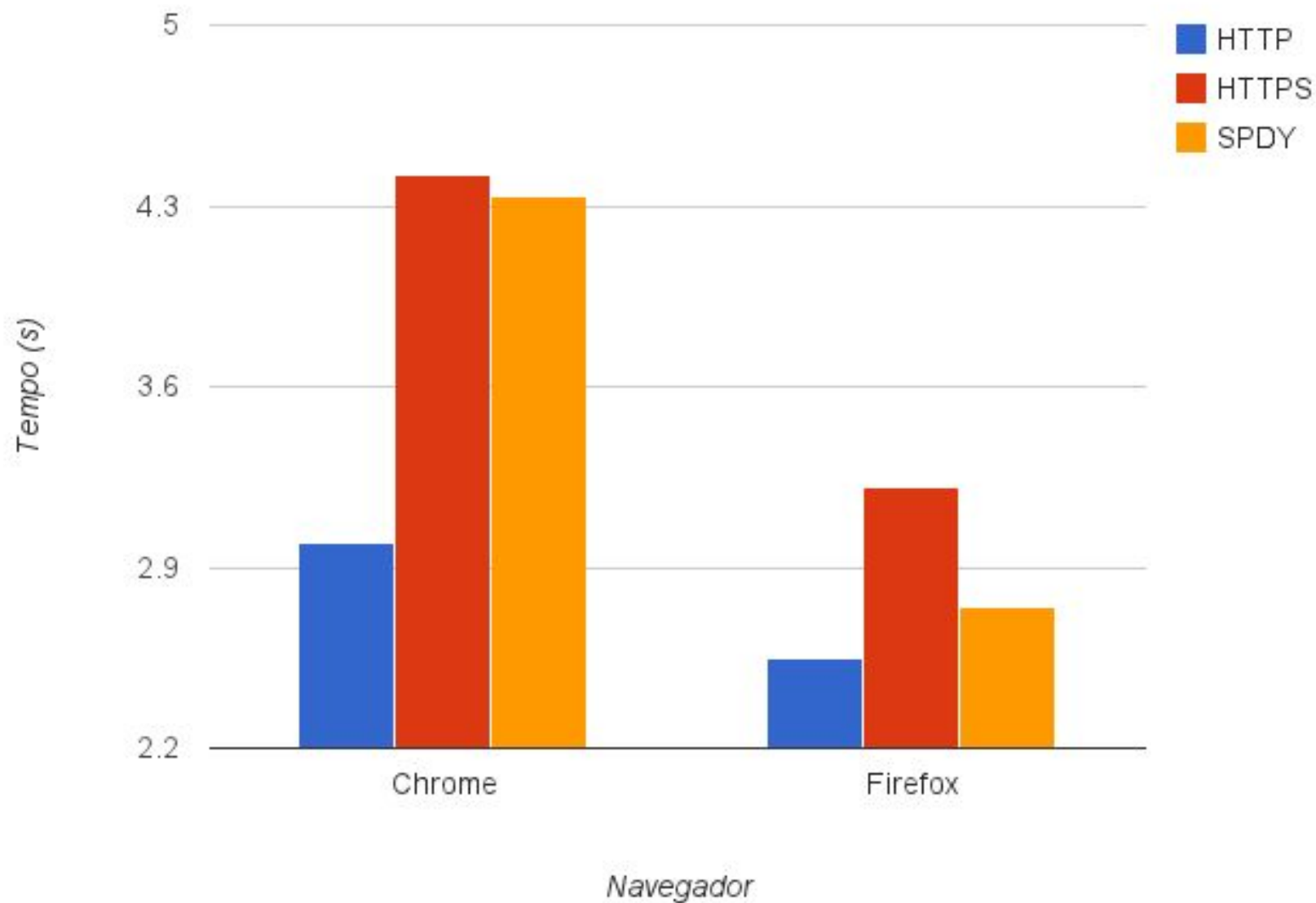
RTT

Banda

Perdas

Browser

Tempo de carregamento (onload) por protocolo



otimizações web http2

diminuir
tráfego



CACHE
MINIFICAÇÃO JS, CSS, HTML
COMPRESSÃO DE IMAGENS

otimizar
renderização



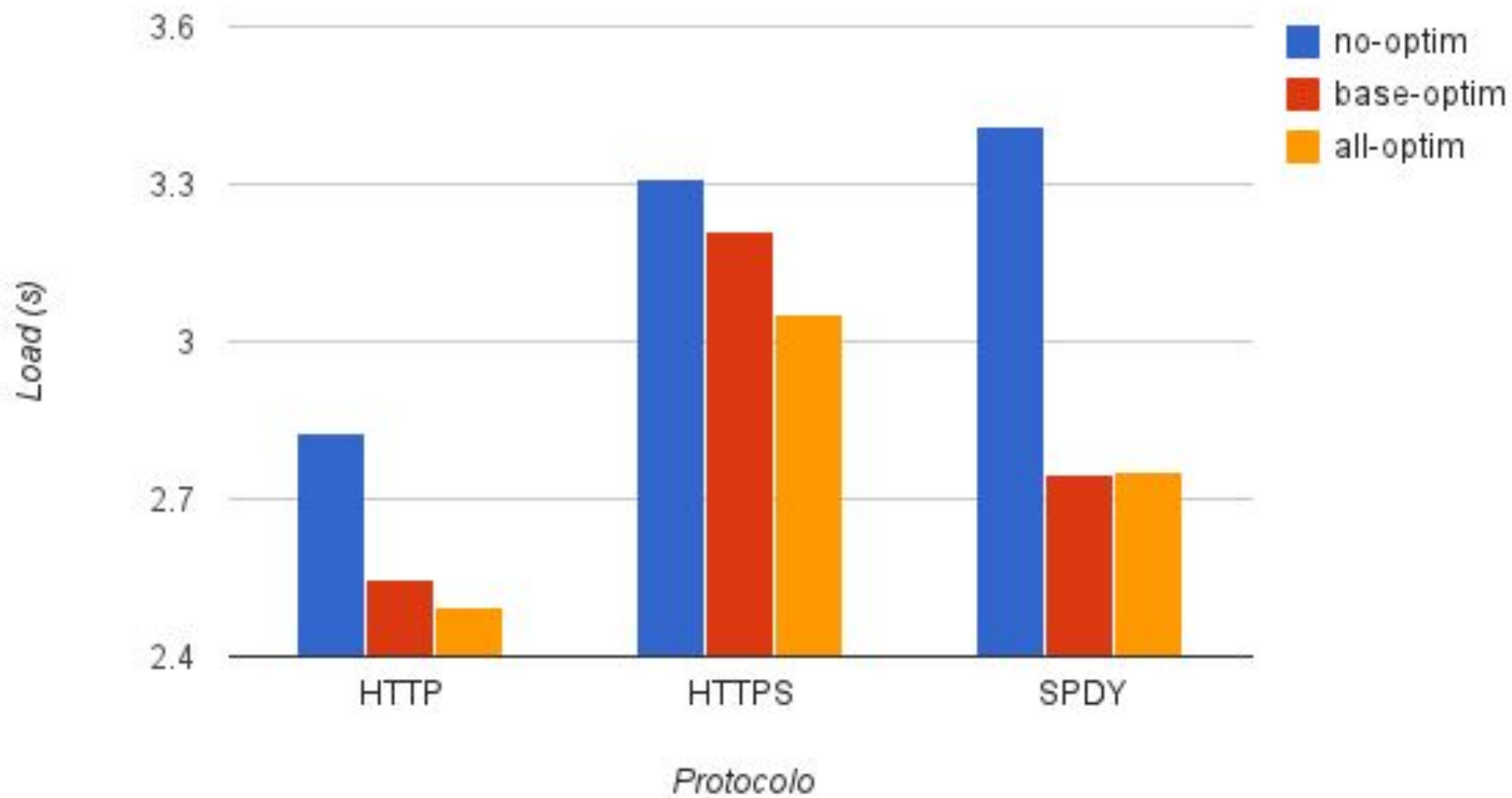
SERVER PUSH / HINT
CRITICAL PATH
60 FPS / JANK FREE

diminuir
latência



CDNs
QUIC

Firefox



http 2

Dez 2013: draft 09 (impl.)

Abr 2014: draft 11

Nov 2014: recomendação

spdy

Nov 2013: draft 3.1

???: draft 4

servidor

nginx

NodeJS

mod_spdy

Jetty

spdy

servidor

nghttp2

NodeJS

Ruby

Go

http 2.0

USE SPDY HOJE

FIQUE DE OLHO NO HTTP2 DE
AMANHÃ

BÔNUS

web components

```
<link rel="stylesheet" href="super-botao.css">  
<template>  
  <button>{{texto}}</button>  
</template>  
<script src="super-botao.js"></script>
```

super-botao.html



```
<link rel="import" href="super-botao.html">
```

web components

BARATO

SERVER PUSH

modularização JS

```
define(["jquery"], function($) {  
    ...  
});
```

home.js

```
define(["jquery"], function($) {  
    ...  
});
```

contato.js

modularização JS

```
define(["jquery"], function($) {  
  ...  
});
```

```
define(["jquery"], function($) {  
  ...  
});
```

home.js

contato.js

LOADER

all.min.js

modularização JS 2.0

```
define(["jquery"], function($) {  
    ...  
});
```

home.js

```
define(["jquery"], function($) {  
    ...  
});
```

contato.js

```
<script src="loader.js"></script>  
<script src="jquery.js" async></script>  
<script src="home.js" async></script>
```

modularização JS 2.0

```
define(["jquery"], function($) {  
    ...  
});
```

home.js

```
define(["jquery"], function($) {  
    ...  
});
```

contato.js

server push



```
<script src="loader.js"></script>  
<script src="jquery.js" async></script>  
<script src="home.js" async></script>
```

modularização

BARATO

CACHEÁVEL

LOADER + SIMPLES

Obrigado!

@srsaude

@sergio_caelum

 caelum
alura

 Casa do Código
Livros e Tecnologia

20% OFF
CUPOM

QCONSP2014

Coletânea Front-end

Uma antologia da comunidade
front-end brasileira



ALMIR FILHO
BERNARD DE LUNA
CAIO GONDIM
DEVID MARQUES
DIEGO EIS
EDUARDO SHIOTA

GIOVANNI KEPPELEN
LUIZ CORTI REAL
JAYDSOM GOMES
REINALDO FERRAZ
SÉRGIO LOPES

 Casa do
Código

links adicionais:

- [The Effect of Network and Infrastructural Variables on SPDY's Performance.](#)
- [Livro High Performance Browser Networking](#)
- [Especificação HTTP2](#)